TEXORA · LECTURE SHEET

Process Control Parameters in Textile Production

Holding each process on target — the parameters that define control in yarn manufacturing, fabric manufacturing, wet processing and apparel manufacturing, how they are measured, and what a control loop does when they drift.

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 7 · LEVEL 4 TERM 1

1. What process control means

Process control is the discipline of keeping a process's measurable characteristics within specified limits, run after run. Every process has **control parameters** — the variables that determine its output quality: some are *set* (draft, temperature, dose), and some are *measured on the output* (count CV %, shade difference, stitches per inch). Control works as a loop: specify the target and tolerance → measure at a defined frequency → compare → correct the setting → verify. A parameter without a target, a tolerance, a measurement method and a correction rule is a hope, not a control.

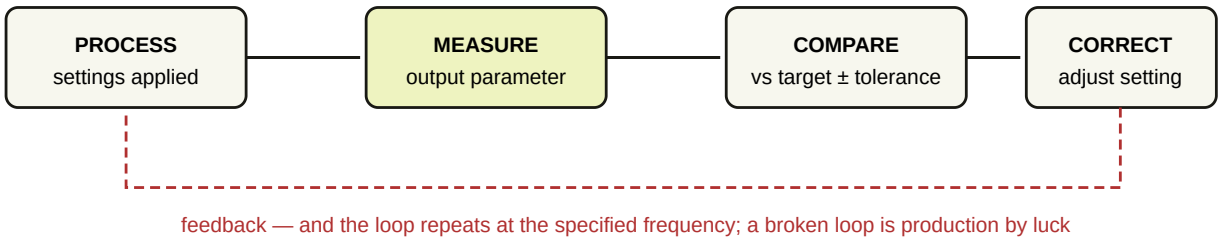


Fig. 1 — The process-control loop. Process control parameters are the quantities this loop measures and corrects.

2. Yarn manufacturing (spinning)

STAGE	CONTROL PARAMETERS	WHY IT MATTERS
Mixing / blowroom	Bale lay-down proportions, trash %, cleaning efficiency, lap/sliver weight per metre and its CV	Errors here cannot be removed later — only diluted
Carding	Sliver hank and CV %, nep count, waste %	Sets the evenness ceiling for the whole yarn

Draw frame / comber	Delivered sliver weight per metre, doubling and draft settings, noil % (comber)	Weight deviation here becomes count deviation in yarn
Ring spinning	Count (and count CV %), twist per unit length, strength (CSP), hairiness, ends-down rate	The saleable properties of the yarn itself
Atmosphere	Department RH and temperature (see TE-4106, Study B)	Regain, breakage and static all follow the air

WORKED EXAMPLE — SLIVER WEIGHT CONTROL TEST

Five consecutive 1-metre sliver cuts from a draw frame weigh: 3.85, 3.92, 3.78, 3.95, 3.90 g.
Mean = (3.85 + 3.92 + 3.78 + 3.95 + 3.90) ÷ 5 = 19.40 ÷ 5 = **3.88 g/m**.
Sample standard deviation: deviations −0.03, +0.04, −0.10, +0.07, +0.02 → squared sum 0.0178 → s = √(0.0178 ÷ 4) = **0.067 g**.
CV % = 0.067 ÷ 3.88 × 100 = **1.72 %**.
Judgement: mean on target (nominal 3.88) and CV within the frame's usual control band — the loop's verdict is "no correction"; a mean shifted by more than the tolerance, or a rising CV, would send the mechanic to draft gears and roller settings respectively. Mean and spread are separate verdicts: one corrects the setting, the other finds the fault.

3. Fabric manufacturing (weaving and knitting)

PROCESS	CONTROL PARAMETERS
Warp preparation	Ends per beam and beam tension uniformity; size add-on %, size viscosity and temperature, moisture in sized warp
Weaving	EPI/PPI (ends and picks per unit length), warp/weft tension, shed timing, loom stop rates by cause (Topic 5), fabric width and GSM at the loom state
Knitting	Course and wale density (CPC/WPC), loop (stitch) length, yarn feed tension, GSM, machine speed and feeder settings
Grey fabric	Width, GSM, defect points per unit area (four-point inspection), skew/bow

Loop length in knitting and pick density in weaving are the master parameters: both fix GSM and dimensional behaviour downstream, and both drift silently (a stretched feed belt, a worn take-up gear) until a fabric lot fails at dyeing or at the buyer's inspection.

4. Wet processing

STAGE	CONTROL PARAMETERS
All baths	Liquor ratio (MLR), temperature profile and time, pH at start and end of dyeing, water hardness
Dyeing	Dye and chemical dosing (g/L and % owf, TE-3106 arithmetic), salt/alkali addition profile, exhaustion and fixation %, shade against standard (ΔE by spectrophotometer), levelness
Finishing	Padder pick-up %, chemical add-on, curing temperature/time, residual formaldehyde and pH of finished fabric where specified
Effluent	pH, BOD/COD, colour, TDS of discharge against consent limits

Wet processing is the least forgiving stage: its parameters interact (temperature accelerates both dyeing and hydrolysis; pH drives both fixation and fibre damage) and its errors are often invisible until the fabric is dry. Hence its control frequency is per-batch, not per-shift.

5. Apparel manufacturing

STAGE	CONTROL PARAMETERS
Cutting	Marker efficiency %, lay height and tension, cut-piece dimensions against pattern, shade/roll separation within a lay
Sewing	Stitches per inch (SPI), seam allowance and seam strength, needle size/type vs fabric, thread tension balance, operation SMVs held (Topic 3)
In-line and end-line quality	DHU (defects per hundred units), defect Pareto by operation, AQL sampling at final audit (as worked in TE-3107/TE-3108)
Finishing/packing	Measurement audit against spec, pressing quality, carton ratio and labelling accuracy

6. Control discipline across all four areas

Three habits separate controlled production from hopeful production. **First, parameter ownership:** every parameter has a named owner, a target, a tolerance, a method and a frequency — written in the department's control plan, not carried in memory. **Second, reaction rules agreed in advance:** when a parameter crosses its limit, the corrective action is already decided (stop, adjust, segregate the lot since the last good check) — improvisation under time pressure is how off-quality lots get shipped. **Third, records that trend:** single readings pass or fail a batch; plotted trends (TE-4108's control charts) warn before limits are crossed and prove stability to buyers. PPC (Topics 1–6) plans the quantities; process control protects the qualities — a schedule met with off-quality goods is not met at all.

Glossary

Control parameter — a measurable variable with target and tolerance that determines output quality. **Control loop** — measure, compare, correct, verify, repeated at a set frequency. **CSP** — count–strength product, the standard yarn strength index. **MLR** — material-to-liquor ratio in wet processing. **ΔE** — numerical colour difference between sample and standard. **DHU** — defects per hundred units inspected. **Reaction rule** — the pre-agreed action when a parameter breaches its limit.

Key points

• A control parameter needs target, tolerance, method, frequency, owner and reaction rule — all six.
• Spinning: sliver weight and CV, count, twist, CSP, ends-down, shed RH — the worked test: mean 3.88 g/m, CV 1.72 %, no correction.
• Fabric: EPI/PPI, loop length, tensions, grey GSM and defect points.
• Wet processing: MLR, temperature–time profile, pH, dosing, exhaustion/fixation, ΔE — controlled per batch.
• Apparel: marker efficiency, SPI, seam quality, DHU and AQL audits.

Practice

1. Four hank readings of roving are 0.92, 0.95, 0.90, 0.93 hank. Compute the mean and comment if the nominal is 0.925 hank with ±2 % tolerance on the mean.
2. For a dyeing department, list six control parameters and for two of them state target-setting source and reaction rule.
3. Explain the difference between correcting for a shifted mean and correcting for a rising CV, using the sliver example.
4. Why is wet-processing control per batch while spinning control can be per shift? Answer in terms of error visibility and lot identity.

Answer key

1. Mean = (0.92 + 0.95 + 0.90 + 0.93) ÷ 4 = 3.70 ÷ 4 = 0.925 hank — exactly nominal; tolerance band on the mean = 0.925 ± 0.0185 = 0.9065–0.9435, so the process is on target and no draft correction is called for. (Individual readings still vary; that is the spread question, separate from the mean.)
2. Parameters (any six): liquor ratio; bath temperature profile; pH; dye dose % owf; salt/alkali dose and addition timing; time at temperature; water hardness; exhaustion %. Example target sources and reaction rules: pH — target from the dye manufacturer's process specification for the dye class in use; rule: if pH is outside the specified band before dye addition, adjust and re-measure before proceeding, never proceed and "correct later". ΔE against standard — target from the buyer's

tolerance (e.g. $\Delta E \leq 1.0$); rule: a batch outside tolerance is held for shading addition or stripping decision by the dye-house manager — it is not passed to finishing on a promise.

3. A shifted mean means the process is centred wrongly: the setting (draft, dose, gear) is adjusted by the size of the shift — a calibration correction. A rising CV at a correct mean means the process has become less consistent — worn rollers, variable material, fluctuating tension — a fault to be found and repaired, not a setting to be nudged. Confusing the two (recentring a process whose spread has exploded) leaves half the output outside tolerance however well the mean is placed.
4. A dye batch is a discrete, identifiable lot whose errors (wrong dose, wrong pH) are committed in one vessel, are usually invisible while wet, and cannot be diluted by later good production — so every batch's parameters are recorded and its result checked before release. Spinning output is a continuous stream from many machines over a shift; its parameters are stable over hours and are verified statistically on samples, because a single bad reading cannot be traced to an isolable "lot" in the same way and the stream is continuously correctable at the next machine setting.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib